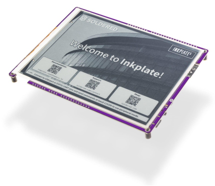
PRODUCT DATASHEET

Inkplate 6

SKU 333232



What it is

Inkplate 6: open-source 6" e-paper display with ESP32, Wi-Fi/Bluetooth, ultra-low power, Arduino/MicroPython ready. Plug-and-play for hobbyists & makers.

Overview

If you're like us, the first time you saw an **e-reader**, you thought to yourself, "I could do something with that." Thanks to clean lines, high contrast, daylight readability, and the remarkable level of **energy efficiency** that comes from drawing power only when changing the contents of the screen, **e-paper** is uniquely suited to many applications. With **Inkplate 6**, our goal is to make e-paper accessible to hobbyists and DIY product designers by offering a plug-and-play hardware platform that is super-easy to use and compatible with **Arduino** and **MicroPython**.

To name a few features, Inkplate 6 has a stunning 6-inch e-paper display with a refresh rate of 1.26 s, partial update 0.26 s, greyscale mode, and partial updates support. Powered by **ESP32**, you will have a strong microcontroller with **Wi-Fi** and **Bluetooth** at your disposal. It's super-low-power (22 µA) so you can use it for days, weeks, or months out of a single battery charge. Using our **Arduino library**, it's up and running in 5 minutes. The **MicroPython module** is available as well. It's 100% open-source for both software and hardware. A **USB-C to USB-C** cable is included in the package, so you can start using it right out of the box.

Designed for high-performance embedded development, the Inkplate 6 fully supports Espressif's official **ESP-IDF** framework. This native environment grants developers low-level control over the hardware, enabling precise memory management, optimized display rendering routines, and custom power profiling to take full advantage of the board's ultra-low deep sleep capabilities. Whether building home prototypes or commercial-grade e-paper applications, ESP-IDF provides the robust foundation needed to unlock the full potential of Inkplate's wireless and processing features.

Inkplate 6 options:

- **With e-paper:** The basic version with all you need to get started.
- **With e-paper display, enclosure, and battery:** Upgrade to this version if you're looking for both protection and mobility. This option includes a 3D-printed enclosure as well as an integrated battery, ensuring your device stays powered even on the go.

Choose the one that best fits your needs and enhance your experience!

Technical details

- **Display:** 6-inch, 800x600 pixel e-paper with greyscale, partial updates, and accelerated refresh cycles
- **Microcontroller:** On-board ESP32 with integrated Wi-Fi and Bluetooth 4.0 (BLE)
- **Power:** Extremely low-energy, battery- or USB-powered operation, including a 25 µA sleep state, with charger
- **Storage:** microSD card reader for image display
- **Design:** Optimized form factor for custom enclosures
- **Clock:** Real Time Clock with battery holder, PCF85063A
- **Connectivity:** Additional GPIO pins, qwiic compatibility, I²C and SPI support
- **Libraries:** Arduino libraries (compatible with Adafruit GFX) and MicroPython module for rendering

- **Enclosure Options:** Optional 3D-printed enclosure
- **Battery Option:** Optional 3D-printed enclosure and 1200 mAh battery
- **Dimensions with EPD:** 144.5 x 107.8 x 10 mm / 5.7 x 4.2 x 0.4 inch
- **Dimensions with Enclosure and Battery:** 160.7 x 116.8 x 13.7 mm / 6.3 x 4.6 x 0.5 inch

Specifications

Display

Display Size	6 inch
Display Resolution	800 × 600 px
Display Color	Grayscale
Frontlight	No
Full Refresh	1.26 s
Partial Refresh	0.26 s
Touch Support	None

MCU

MCU Part Number	ESP32-WROVER-E
MCU Clock	240 MHz

Connectivity

Wi-Fi	Wi-Fi 4
Bluetooth	Classic + BLE

Interface

qwiic Compatible	Yes
SD Card Slot	Yes

Power

Battery Included	No
Battery Connector	JST 2-pin
Sleep Current	25 µA

Other

Dev Environments	Arduino IDE, MicroPython, ESP-IDF, ESPHome / Home Assistant
Has Enclosure	No
RTC Onboard	Yes

Typical applications

Inkplate 6 is a compact e-paper display module for connected dashboards, price tags and environmental monitors. It suits projects where you want a low-power screen that stays readable without a backlight. Connect it to your ESP32 or Arduino board, then update the content over Wi-Fi for a display that can run for days on a single charge.

What’s in the box

- 1 × Inkplate 6
- 1 × USB-C Cable

Variants

- With e-paper — SKU 333232
- With e-paper, Enclosure & Battery — SKU 333229

Resources

Technical Resources	Hardware Details WEB
Documentation	Full Documentation DOCS
Software Resources	Arduino Library GITHUB · Arduino: Get Started (docs.soldered.com) DOCS · Arduino: Get Started (inkplate.readthedocs.io) DOCS · MicroPython Module GITHUB
Compliance	OSHWA Certification PDF

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